



SIMATS ENGINEERING

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Course Code: CSA1218

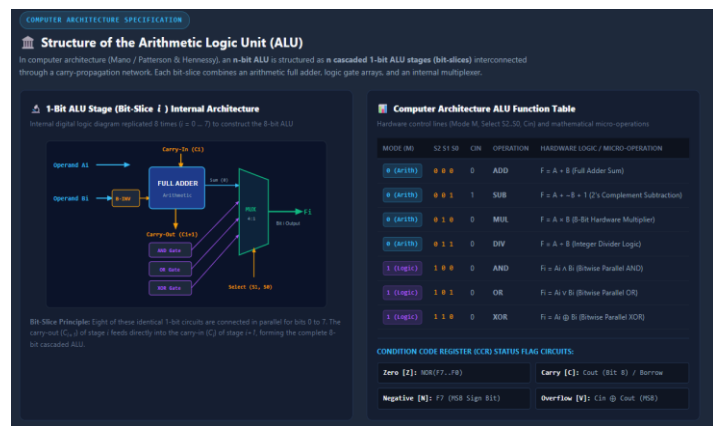
Slot: B

Course Name: COMPUTER ARCHITECTURE

Course Faculty: Dr. K. JAYASAKTHI VELMURUGAN

Project Title: DESIGN AND SIMULATION OF AN ARITHMETIC LOGIC UNIT FOR PROCESSOR SIMULATIONS

Module Photographs:



Project Description:

The **Web-Based Simulation of an 8-bit RISC Processor with ALU** is an interactive system that demonstrates the working of a processor and its ALU operations. It simulates instruction execution through the **Instruction Memory, Decoder, Control Unit, Register File, ALU, Flags, and Result Register**. The system automatically assigns registers and operand values and performs **ADD, SUB, MUL, DIV, AND, OR, and XOR** operations. Real-time datapath animations show the flow of data during execution, while processor flags and results are displayed instantly. The project also provides graphical performance analysis using **Chart.js**, comparing **execution time, computational complexity, and relative efficiency** of different ALU operations. Overall, the project combines **processor architecture, ALU design, software simulation, real-time visualization, and performance analysis** into a single web-based platform. It helps students understand the internal working of an 8-bit RISC-style processor and provides a practical demonstration of concepts that are otherwise difficult to visualize through theoretical study alone.

Student Signature

Guide Signature